

Scenario 1

Scenario 1 – Department Salary Benchmark Analysis

The HR department wants to identify employees earning more than the average salary of their department. Include only departments whose average salary is greater than the overall company average salary.

Display:

- Employee ID
- Employee Name
- Department
- Employee Salary
- Department Average Salary
- Company Average Salary

Sort:

1. Department (Ascending)
2. Employee Salary (Descending)

SQL Solution

```
WITH DeptAvg AS
(
    SELECT Department,
           AVG(Salary) AS Dept_Avg_Salary
    FROM Employee
    GROUP BY Department
),
CompanyAvg AS
(
    SELECT AVG(Salary) AS Company_Avg_Salary
    FROM Employee
)
SELECT E.Emp_ID, E.Emp_Name, E.Department,
       E.Salary AS Employee_Salary,
       D.Dept_Avg_Salary,
       C.Company_Avg_Salary
FROM Employee E
JOIN DeptAvg D ON E.Department=D.Department
CROSS JOIN CompanyAvg C
WHERE E.Salary>D.Dept_Avg_Salary
      AND D.Dept_Avg_Salary>C.Company_Avg_Salary
ORDER BY E.Department ASC, E.Salary DESC;
```

Scenario 2

Scenario 2 – Top Performing Customers Analysis

Display customers whose:

- Total purchase amount is above the average of all customers.
- Total orders are greater than 3.
- Latest order was placed in the current year.

Display:

- Customer ID
- Customer Name
- Total Orders
- Total Purchase Amount
- Latest Order Date

Sort:

1. Total Purchase Amount (Descending)
2. Total Orders (Descending)

SQL Solution

```
WITH CustomerSummary AS
(
    SELECT C.Customer_ID, C.Customer_Name,
           COUNT(O.Order_ID) AS Total_Orders,
           SUM(O.Order_Amount) AS Total_Purchase_Amount,
           MAX(O.Order_Date) AS Latest_Order_Date
    FROM Customer C
    JOIN Orders O ON C.Customer_ID=O.Customer_ID
)
```

```

        GROUP BY C.Customer_ID, C.Customer_Name
    ),
    AveragePurchase AS
    (
        SELECT AVG(Total_Purchase_Amount) AS Avg_Total_Purchase
        FROM CustomerSummary
    )
    SELECT CS.Customer_ID, CS.Customer_Name,
           CS.Total_Orders,
           CS.Total_Purchase_Amount,
           CS.Latest_Order_Date
    FROM CustomerSummary CS
    CROSS JOIN AveragePurchase AP
    WHERE CS.Total_Purchase_Amount>AP.Avg_Total_Purchase
           AND CS.Total_Orders>3
           AND YEAR(CS.Latest_Order_Date)=YEAR(CURDATE())
    ORDER BY CS.Total_Purchase_Amount DESC,
           CS.Total_Orders DESC;

```